

$$\equiv x_i + 20 + \left(5\left(\frac{1^2}{x}\right) + 0.00\right) \frac{1.2}{h \bar{c}_j}$$

$$f(x + (5x =, x(0) + = 10, = + x_i^-)$$

$$2\left(\frac{v}{xt}\right)^{\frac{2}{2}} + (x)) ++ \left(\frac{2x}{x}\right) = I_{42}((x) = \frac{2^2}{h} = 5_x^{\frac{2}{x}}$$

$$\frac{x(1x \frac{2}{x} \frac{2}{2}}{h x 3}$$

$$f(\bar{x}_2 =$$

$$x_{x-} + x(x_{x} + x + \left(\frac{kx}{m_{12}^2}\right) \mp 1$$

$$= \frac{x}{x_j} + 0 - (x_{x-} + 0) = \frac{x}{l_x am^{x_x}}$$

$$f + (0, + 5_{j_i} + x_i = (1_i = 3$$

Engineering Mathematics III:

- Eigenvalues and Eigenvectors
- Systems of Differential Equations
- Converting a 2nd order ODE into a system of 1st order ODEs

Week 5 Lecture 5

Housekeeping

1 Pre-class recordings

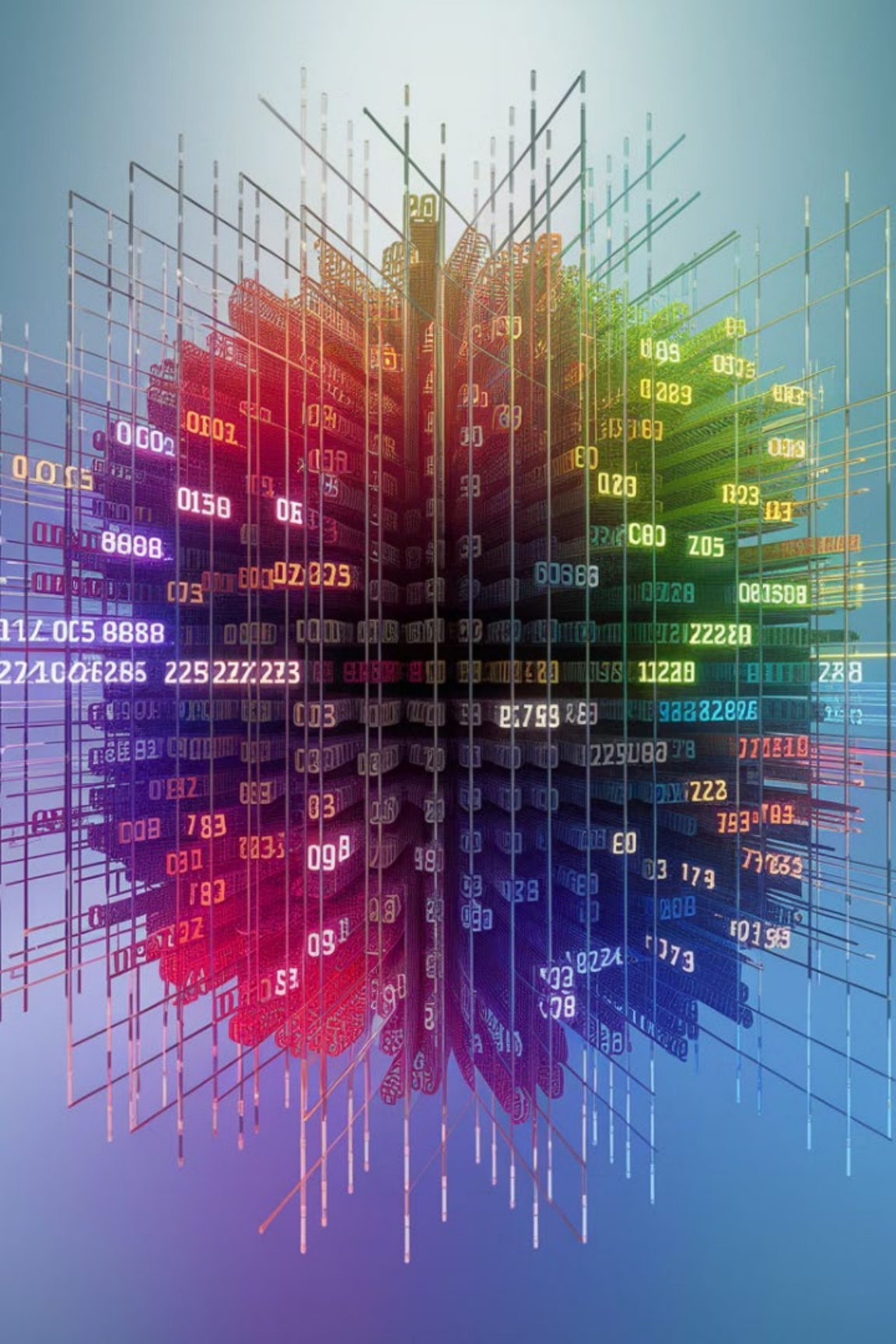
Available on Canvas - access both sections

2 Assignment 1

Quizzes 1 to 6 : Due 21st April

Assignment 1 Part B released this Friday: Due April 27th 2025





Matrix Mastery: Preparing for Eigenvalues & Eigenvectors

Essential matrix operations and concepts.

What Are Matrices?

$$\begin{pmatrix} 2 & -1 & 0 \\ 3 & \sqrt{2} & \frac{1}{2} \end{pmatrix} \quad \begin{pmatrix} 9 & -2 & 0 \\ 2 & 1 & -3 \\ 5 & -1 & 2 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 6 \\ -1 \end{pmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Definition

A *matrix* is a rectangular array of
of (real) numbers arranged in
rows (n) and columns (w).

Order of a *matrix* = n x m

Special Types

Square: equal rows/columns

Diagonal: non-zero entries only on
only on diagonal

Identity: 1s on diagonal,
0s elsewhere

Operations of Matrices

Addition: If the matrices are the same order, add corresponding elements

Multiplication: A matrix can be multiplied by a scalar.

Two Matrices (C and D) can be multiplied together if the number of columns in C equals the number of rows in D , using the dot product of each row and column.

For the following matrices, find

a) $C + D$

b) $3D$

c) $2C - 3D$

$$C = \begin{bmatrix} 4 & -1 \\ 0 & 4 \\ -3 & 8 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 4 \\ -1 & 3 \\ 1 & 7 \end{bmatrix}$$

If possible, find

a) EF

b) FE

$$E = \begin{bmatrix} 4 & -1 & 4 & -1 \\ 2 & -1 & 2 & 3 \end{bmatrix} \quad F = \begin{bmatrix} 6 & 4 \\ 2 & -1 \end{bmatrix}$$

Order matters when multiplying matrices!

Transpose

The **transpose** of an $m \times n$ matrix A is the $n \times m$ matrix A^T obtained by interchanging the rows and columns of A so the i, j -entry of A becomes the j, i -entry of A^T

Example 6

$$P = \begin{pmatrix} 4 & -3 & 2 \\ 0 & 1 & -1 \end{pmatrix} \qquad P^T = \begin{pmatrix} & \\ & \\ & \end{pmatrix}$$

$$P = \begin{pmatrix} 4 & -3 & 2 \\ 0 & 1 & -1 \end{pmatrix} \qquad P^T = \begin{pmatrix} 4 & 0 \\ -3 & 1 \\ 2 & -1 \end{pmatrix}$$

2×3 3×2

Inverse matrix for a 2×2 matrix A

The following procedure is used to calculate the inverse of a 2×2 matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \qquad A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

First find the number $a_{11}a_{22} - a_{12}a_{21}$

This number is called the **determinant** $\det(A)$ of the matrix A

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Example 2

Use the procedure to find the inverse matrix for $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$

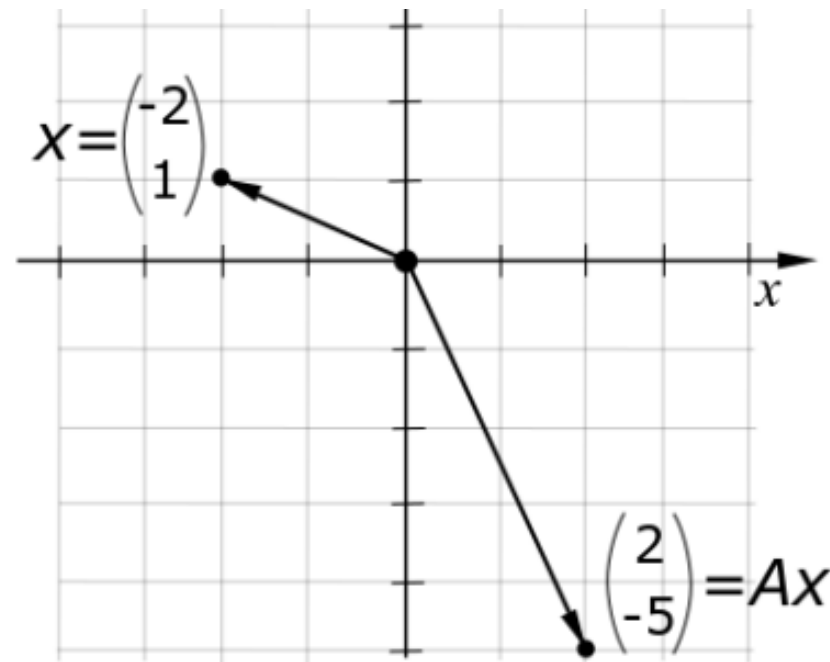
$$\det(A) =$$

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} =$$

$$\begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

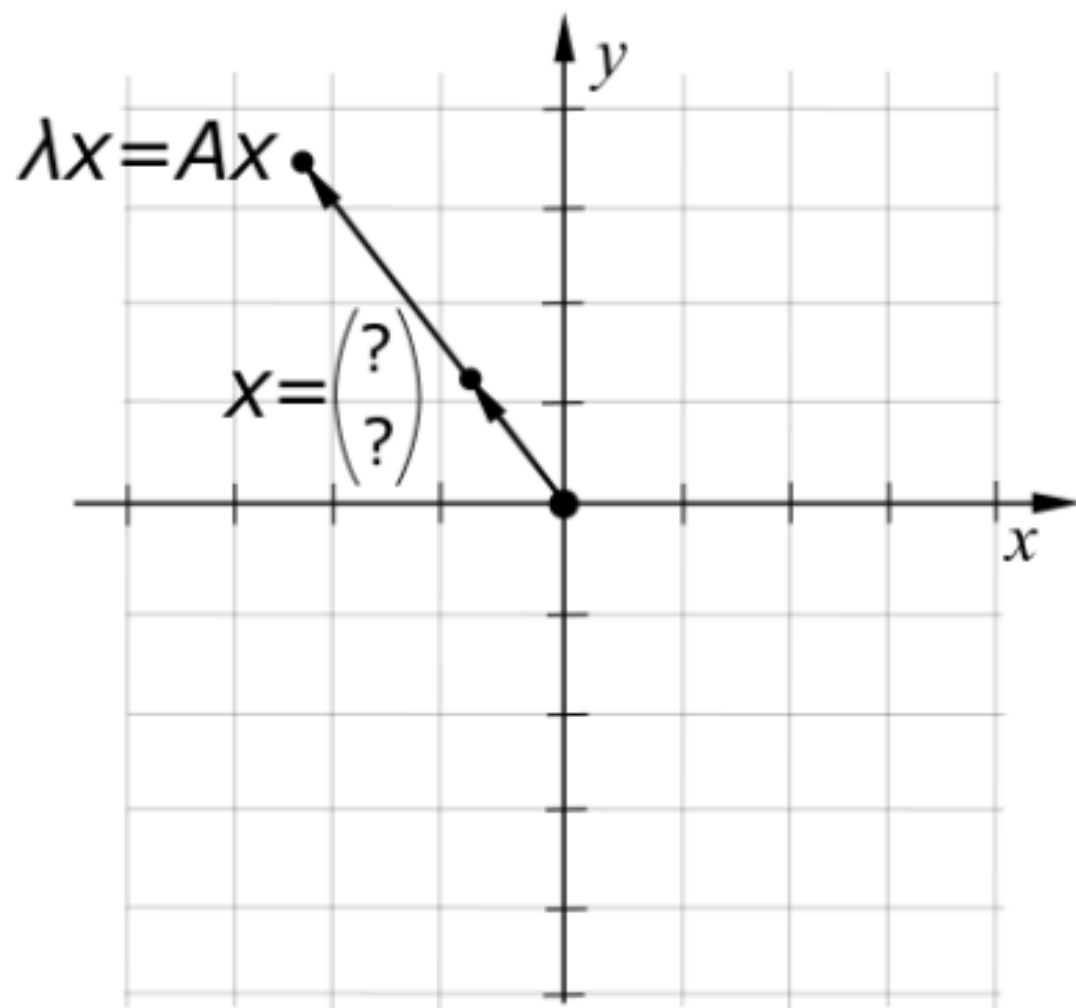
More generally, a vector multiplied by a matrix will give another vector. In this way we can view matrices as functions on a vector space: they take a vector as input, and give another vector as output. It turns out these represent a special kind of function called a **linear function**.

Lets look at what the input and output of our example look like on a graph:



We input a $x = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, and get vector $Ax = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ as an output.

Question: with our matrix $A \begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix}$, can we find a vector x such that Ax is a scalar multiple of x ?



This gives the **eigenvector problem**. It can be stated as follows: given a matrix A , find a vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

Such a vector $\mathbf{x}$ is called an **eigenvector**, and the corresponding value λ is its **eigenvalue**.

Finding eigenvalues and eigenvectors

First we need to calculate the eigenvalues. These are found by solving

$$\det(A - \lambda I) = 0.$$

This just means subtracting λ from the diagonal of A . This is best illustrated in our example.

We have

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 4 \\ 2 & -1 - \lambda \end{pmatrix}.$$

We have

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 4 \\ 2 & -1 - \lambda \end{pmatrix}.$$

Then

$$\begin{aligned} \det(A - \lambda I) &= (1 - \lambda)(-1 - \lambda) - (2)(4) \\ &= \lambda^2 - 1 - 8 \\ &= \lambda^2 - 9 \end{aligned}$$

so $\lambda = 3$ and $\lambda = -3$ are our two eigenvalues.

Now we can find our eigenvectors. There will be one for each eigenvalue. We can find them as follows:

For an eigenvalue λ , we need to find a vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ such that

$$(A - \lambda I)\mathbf{x} = \mathbf{0}.$$

Since we have already found our eigenvalues λ , we have that $(A - \lambda I)$ will just be a matrix with numbers. Lets see how to do this with our example.

For $\lambda = 3$:

We have

$$A - \lambda I = \begin{pmatrix} 1 - 3 & 4 \\ 2 & -1 - 3 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 2 & -4 \end{pmatrix}.$$

We need to find a vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ such that

$$\begin{pmatrix} -2 & 4 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We need to find a vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ such that

$$\begin{pmatrix} -2 & 4 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

or

$$-2x_1 + 4x_2 = 0$$

$$2x_1 - 4x_2 = 0$$

This gives two equations and two unknowns, but (if we have done this right) one equation is a scalar multiple of the other, so there is no unique solution.

The easiest way to proceed is to choose a (non-zero) value for one of the variables (for example $x_1 = 1$), and work out what the other variable will be. The solution we get should work for both equations.

In this case we let $x_1 = 1$, so then

$$\begin{aligned} (-2)(1) + 4x_2 &= 0 \\ x_2 &= \frac{1}{2} \end{aligned}$$

So $x_1 = 1$ and $x_2 = \frac{1}{2}$

Our first eigenvector corresponding to the eigenvalue $\lambda = 3$ is then $\begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}$.

(note this is not unique - any scalar multiple of $\begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}$ will also be an eigenvector)

Exercise: Find an eigenvector corresponding to the eigenvalue $\lambda = -3$.

See Differential Equations 3rd edition by Blanchard et al. pages 262-267 for more information on how to calculate eigenvalues and eigenvectors.

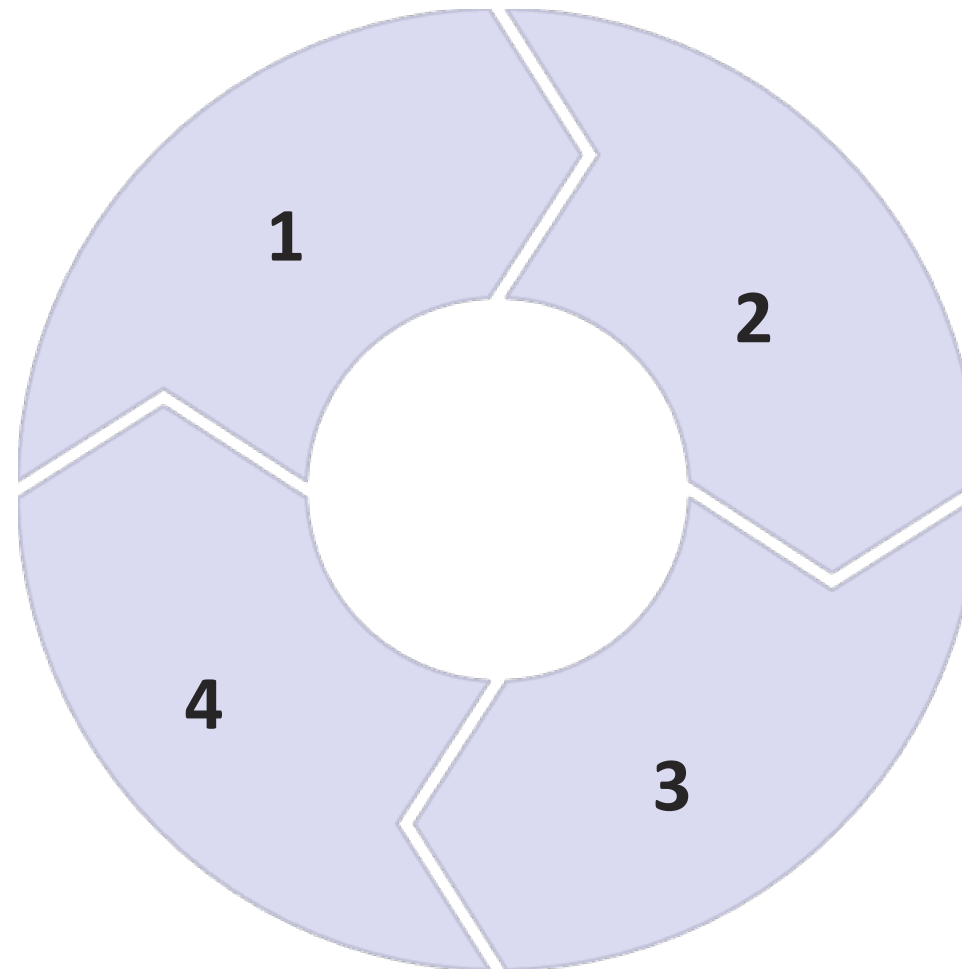
Eigenvalues & Eigenvectors

Definition

$Av = \lambda v$ where v is eigenvector, λ is eigenvalue

Finding Eigenvectors

Substitute λ into $(A - \lambda I)v = 0$ and solve
solve



Characteristic Equation

$$\det(A - \lambda I) = 0$$

Finding Eigenvalues

Solve polynomial equation for λ



Practice Problems

Problem	Answer
Find eigenvalues of $\begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$	Eigenvalues 3 and -2 with eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Find eigenvectors for $\begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}$	Eigenvalues -1 and 5 with eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Diagonalize $\begin{bmatrix} 3 & 2 \\ 0 & -2 \end{bmatrix}$	Eigenvalues -2 and 3 with eigenvectors $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Systems of Differential Equations

Now we will look at **systems of linear differential equations** with constant coefficients.

In this course we will only consider systems with two equations.

They will be of the form:

$$\begin{aligned}y_1' &= a_{11}y_1 + a_{12}y_2 \\ y_2' &= a_{21}y_1 + a_{22}y_2\end{aligned}$$

Note that this can be written in the form:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Which can also be written as:

$$y' = Ay$$

Finding Solutions to Systems of Differential Equations

1

Start with the System

We have a system of differential equations

$$\mathbf{y}' = A\mathbf{y}$$

This looks similar to the differential equation

$$y' = ky$$

for which the solution was

$$y = Ce^{kt}$$

2

Assume a Solution Form

We will assume a solution to the system is of the form

$$\mathbf{y} = \mathbf{x}e^{\lambda t}$$

for some vector $\mathbf{x}$ and some constant λ . Then

$$\mathbf{y}' = \lambda \mathbf{x}e^{\lambda t},$$

3

Derive the Condition

but we also know from the formulation of our problem that $\mathbf{y}' = A\mathbf{y}$, so we put that together with $\mathbf{y} = \mathbf{x}e^{\lambda t}$, and we have

$$\mathbf{y}' = A\mathbf{x}e^{\lambda t}$$

So

$$A\mathbf{x}e^{\lambda t} = \lambda\mathbf{x}e^{\lambda t}$$

so we need a constant λ and a vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}.$$

$$\mathbf{y}' = A\mathbf{y}$$

$$\mathbf{y} = \mathbf{x}e^{\lambda t}$$

$$\mathbf{y}' = \lambda\mathbf{x}e^{\lambda t}$$

4

Find Eigenvalues and Eigenvectors

This is exactly the process of finding eigenvalues and eigenvectors of A !

Our general solution will have two solutions derived from the two eigenvalues and their associated eigenvectors.

$$\mathbf{y} = c_1\mathbf{x}_1e^{\lambda_1 t} + c_2\mathbf{x}_2e^{\lambda_2 t}$$

where $\mathbf{x}_1 = \begin{pmatrix} x_{11} \\ x_{21} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} x_{12} \\ x_{22} \end{pmatrix}$ are eigenvectors of matrix A

corresponding to eigenvalues λ_1 and λ_2 respectively, and c_1 and c_2 are arbitrary constants (which we could find if we had initial conditions).

Find Eigenvalues and Eigenvectors

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where $\mathbf{x}_1 = \begin{pmatrix} x_{11} \\ x_{21} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} x_{12} \\ x_{22} \end{pmatrix}$ are eigenvectors of matrix A corresponding to eigenvalues λ_1 and λ_2 respectively, and c_1 and c_2 are arbitrary constants (which we could find if we had initial conditions).

This vector equation can then read:

$$y_1 = c_1 x_{11} e^{\lambda_1 t} + c_2 x_{12} e^{\lambda_2 t}$$

$$y_2 = c_1 x_{21} e^{\lambda_1 t} + c_2 x_{22} e^{\lambda_2 t}$$

Example: Find the general solution to the following system of differential equations:

$$y_1' = 7y_1 - y_2$$

$$y_2' = 3y_1 + 3y_2$$

We can express

$$\begin{aligned}y_1' &= 7y_1 - y_2 \\y_2' &= 3y_1 + 3y_2\end{aligned}$$

as

$$\mathbf{y}' = \begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix} \mathbf{y}.$$

Now we need to find the eigenvalues and eigenvectors of the matrix

$$\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix}$$

Start by solving $\det(A - \lambda I) = 0$.

$$(7 - \lambda)(3 - \lambda) + 3 = 0$$

$$\lambda^2 - 10\lambda + 24 = 0$$

$$\lambda = 6 \quad \text{or} \quad \lambda = 4$$

Now we can find the eigenvectors:

For $\lambda = 6$:

$$\begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $x_1 = 1$ then $x_2 = 1$ will work.

So $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is our first eigenvector.

For $\lambda = 4$:

$$\begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $x_1 = 1$ then $x_2 = 3$ will work. So $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ is our second eigenvector

Our general solution will then be:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}$$

$$\mathbf{y} = c_1 \mathbf{x}_1 e^{\lambda_1 t} + c_2 \mathbf{x}_2 e^{\lambda_2 t}$$

Example: Solve the initial value problem:

$$y_1' = 7y_1 - y_2$$

$$y_2' = 3y_1 + 3y_2$$

where $y_1(0) = 1$, $y_2(0) = 0$.

This gives us

$$1 = c_1 + c_2$$

$$0 = c_1 + 3c_2$$

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}$$

<https://www.youtube.com/watch?v=-HBLJ3mm6CY>

Initial value problem for system of 1st order differential equations

This gives us

$$\begin{aligned}1 &= c_1 + c_2 \\ 0 &= c_1 + 3c_2\end{aligned}$$

Solving this system of equations gives $c_1 = \frac{3}{2}$ and $c_2 = -\frac{1}{2}$, so our particular solution becomes:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} - \frac{1}{2} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}.$$

<https://www.youtube.com/watch?v=-HBLJ3mm6CY>

Initial value problem for system of 1st order differential equations

Exercise: Solve the following initial value problem:

$$\begin{aligned}y_1' &= y_1 - y_2 \\ y_2' &= 2y_1 - 2y_2,\end{aligned}$$

where $y_1(0) = 1$, $y_2(0) = 0$.

Example: Solve the initial value problem:

$$\begin{aligned}y_1' &= 7y_1 - y_2 \\ y_2' &= 3y_1 + 3y_2\end{aligned}$$

where $y_1(0) = 1$, $y_2(0) = 0$.

To solve an initial value problem we would need to first find the general solution, and then apply the initial conditions to get values for the constants.

In this case the system is the same as the last one, so we can just use the general solution we just found:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}.$$

We know that $y_1(0) = 1$ and $y_2(0) = 0$, applying this we get:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{(6)(0)} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{(4)(0)}.$$

Example: Two tanks each initially hold 100 litres of fluid. The fluid in tank 1 is pure water, whereas 30kg of salt has been dissolved in pure water in tank 2.

The solutions are exchanged between the tank at a rate of 5 litres per minute (assume the tanks are stirred so the solutions are uniform in both tanks).

Let $y_1(t)$ describe the amount of salt in tank 1 at time t and $y_2(t)$ describe the amount of salt in tank 2 at time t .

How long until there is half as much salt in tank 1 as there is in tank 2?

First we must set up a model of the system. We have:

$$y_1' = \text{Inflow} - \text{Outflow} = \frac{5}{100}y_2 - \frac{5}{100}y_1$$

$$y_2' = \text{Inflow} - \text{Outflow} = \frac{5}{100}y_1 - \frac{5}{100}y_2$$

This gives a system of differential equations:

$$y_1' = -0.05y_1 + 0.05y_2$$

$$y_2' = 0.05y_1 - 0.05y_2,$$

where $y_1(0) = 0$, $y_2(0) = 30$

This gives a system of differential equations:

$$\begin{aligned}y_1' &= -0.05y_1 + 0.05y_2 \\ y_2' &= 0.05y_1 - 0.05y_2,\end{aligned}$$

where $y_1(0) = 0$, $y_2(0) = 30$

Exercise: Solve this initial value problem and answer the original question.

(hint: there will be half as much salt in tank 1 as there is in tank 2 when tank 1 contains $\frac{1}{3}$ the total amount of salt).

We will now show how to convert a second order homogeneous differential equation with constant coefficients to a system of two first order differential equations with constant coefficients.

Suppose we have a differential equation

$$ay'' + by' + cy = 0.$$

The trick to converting it comes from defining the vector

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} y \\ y' \end{pmatrix}.$$

Then

$$\mathbf{v}' = \begin{pmatrix} v_1' \\ v_2' \end{pmatrix} = \begin{pmatrix} y' \\ y'' \end{pmatrix}.$$

Without loss of generality, assume $a = 1$. Note that rearranging our second order equation we get

$$y'' = -cy - by'.$$

$$ay'' + by' + cy = 0.$$

We can then set up the system

Show how the system is setup

$$\mathbf{v}' = \begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} y' \\ -cy - by' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix} = A\mathbf{v}$$

In this case the top row of the matrix A is always $0 \ 1$, and the bottom row coefficients come from the original second order equation.

A second order equation: $y'' + by' + cy = 0$

$$\Rightarrow y'' = -by' - cy.$$

Let $y_1 = y$, $y_2 = y'$.

$$y' = y' \Rightarrow y_1' = y_2$$

$$\Rightarrow \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$(y')' = -by' - cy \Rightarrow y_2' = -by_2 - cy_1$$

$$\mathbf{v}' = \begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} y' \\ -cy - by' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix} = A\mathbf{v}$$

In this case the top row of the matrix A is always $0 \quad 1$, and the bottom row coefficients come from the original second order equation.

If we have initial conditions $y(0) = \alpha$ and $y'(0) = \beta$, we can apply these too, and get

$$\mathbf{v}(0) = \begin{pmatrix} y(0) \\ y'(0) \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Example: Solve the initial value problem

$$y'' + 5y' + 4y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

by converting it to a system of two first order differential equations.

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by converting it to a system of two first order differential equations.

We set

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} y \\ y' \end{pmatrix},$$

and get

$$\mathbf{v}' = \begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} y' \\ -4y - 5y' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -4 & -5 \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix}$$

with initial condition

$$\mathbf{v}(0) = \begin{pmatrix} y(0) \\ y'(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Exercise: Solve this initial value problem.

$$\mathbf{v}' = \begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} y' \\ -cy - by' \end{pmatrix}$$

In this case the top row of the matrix A is always $0 \ 1$, and the bottom row coefficients come from the original second order equation.

If we have initial conditions $y(0) = \alpha$ and $y'(0) = \beta$, we can apply these too, and get

$$\mathbf{v}(0) = \begin{pmatrix} y(0) \\ y'(0) \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Solve an initial value problem

1. Convert 2nd order ODE into system of 2 1st order ODEs.
2. Find the general solution. (Find eigenvalues and eigenvectors for system matrix).

$$\mathbf{y} = c_1 \mathbf{x}_1 e^{\lambda_1 t} + c_2 \mathbf{x}_2 e^{\lambda_2 t}$$

3. Apply the initial conditions to get values for the constants.
4. Sub back into general solution to find particular solution.

Exercises

- Find the general solution for the following system of equations:

$$\begin{aligned}y_1' &= y_1 - 3y_2 \\ y_2' &= 0y_1 - 2y_2\end{aligned}$$

- Solve the initial value problem

$$y'' - 3y' + 2y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

by converting it to a system of two first order equations.

To do Week 5:

- Watch ENGE702 lecture 12 pre-class recording on Canvas
- Watch ENGE702 lecture 13 pre-class recording on Canvas
- Watch ENGE702 lecture 14 pre-class recording on Canvas
- Complete Quiz 5
- Read Textbook Advanced Engineering Textbook 4.0, 4.1
- Review tutorial problems before this week's tutorial
- Matlab- 'MATLAB Onramp' COMPLETED
Work on 'Solving Ordinary Differential Equations With MATLAB'



Textbook References

Chapters: 4.0, 4.1

Bryan (2022) Chapter 6



What did you get out of today? today?

- **Purpose of today's lecture?** What was the purpose of today's lecture?
- **Key learning:** What is one key thing you learnt?
- **Questions:** Is there anything that is still puzzling you?

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